

Kemoplat Inj. - PIL - MYS - Baddi-2

For the use only of a Registered Medical Practitioner
or a Hospital or a Laboratory

Kemoplat Injection 1 mg/ml

KEMOPLAT

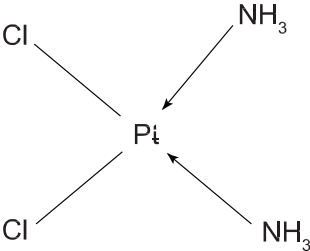
DESCRIPTION

Cisplatin is a cell cycle-phase nonspecific anti-neoplastic agent used as an adjunct in the treatment of a wide variety of tumours. Cisplatin is an inorganic compound closely related to alkylating agents such as cyclophosphamide. Cisplatin and Carboplatin have identical mechanism of action and differ only in their structures and toxicities. KEMOPLAT (Cisplatin Injection) is a clear, colourless solution.

COMPOSITION

KEMOPLAT (Cisplatin Injection 1.0 mg/ml)
Each ml contains:
Cisplatin USP 1.0 mg
Sodium Chloride USP 9 mg
Hydrochloric Acid USNF q.s. to adjust pH
Water for injection USP q.s.

CHEMICAL STRUCTURE



Cisplatin is a heavy metal platinum co-ordination complex containing a central atom of platinum surrounded by two entities each of two chloride ions and two ammonia molecules in Cis-geometry. Chemically, it is a cis isomer of diamminedichloroplatinum (II). It has a melting point of 207°C. The empirical formula of the active compound is PtCl₂ H₂ N₄ with a molecular weight of 300.1.

PHARMACOLOGY

MECHANISM OF ACTION¹

Cisplatin is an antineoplastic agent with biochemical properties similar to those of bifunctional alkylating agents. The drug covalently binds to DNA bases and disrupts DNA function. Cisplatin appears to enter cells by diffusion. Hydrolysis of chloride in its structure is responsible for formation of activated species of the drug, which reacts with nucleic acids and protein. Cisplatin exerts its action by forming both intrastrand and interstrand crosslinks with complementary strands of DNA, thus inhibiting DNA synthesis. Protein and RNA synthesis is also inhibited, but to a lesser extent. Its antineoplastic actions are not cell-cycle phase specific.

PHARMACOKINETICS²

Following intravenous administration, Cisplatin is rapidly distributed to all tissues, except the central nervous system, where it does not penetrate to an appreciable extent. The highest concentrations are present in liver, kidneys, bladder, muscle, skin, testis and prostate. Cisplatin has a volume of distribution of 0.17-1.47 L/kg. After intravenous administration, plasma

elimination is biphasic with half-lives of 25-49 minutes (Phase-I) and 58-73 hour (Phase-II), respectively. Cisplatin is eliminated via the urine (27 to 43%) of the administered dose is excreted in the urine during the first five days after administration.

DRUG INTERACTIONS^{3,4}

- Potentially nephrotoxic or ototoxic drugs, e.g. aminoglycoside, antibiotics and loop diuretics may potentiate the nephrotoxic and ototoxic effects of Cisplatin.
- Cisplatin by reducing the renal function decreases the renal clearance of renally excreted drugs (especially ifosfamide, high dose methotrexate) and increases the toxicities of these drugs by increasing their half-life.
- Cisplatin decreases phenytoin serum level by decreased absorption and/or increased metabolism of phenytoin.
- Cisplatin when used along with etoposide has a synergistic effect against certain tumours (testicular cancer, lung cancer) possibly by decreasing the clearance of etoposide.

INDICATIONS

- Metastatic, non-seminomatous germ cell carcinoma
- Advanced stage and refractory ovarian carcinoma
- Advanced stage and refractory bladder carcinoma

CONTRAINDICATIONS

Absolute

In patients with a history of hypersensitivity to Cisplatin or other platinum containing compounds.
It should not be used in pregnancy or lactation.

Relative

Since it induces nephrotoxicity, myelosuppression and ototoxicity and may act as additive preexisting dysfunction, its use in renal or hearing impairment or depressed bone marrow may be associated with enhanced toxicities. Therefore they represent relative contraindication.

ADVERSE REACTIONS^{7,8}

Clinical use of Cisplatin may be limited because of the number of drug induced toxic reactions, their time course and frequency generally depending upon dose schedule and drug delivery. Clinical predisposing factors may include hyperuricaemia & hypoalbuminemia

Nephrotoxicity

Cumulative and dose related renal impairment is the major dose limiting adverse reaction. Renal toxicity has been noted in 28 to 36% of patients treated with a single dose of 50 mg/m². Renal toxicity can be divided into two phases-acute & chronic.

Acute phase: It is observed usually during first 24-48 hrs. especially in patients receiving inadequate pretreatment hydration. It is characterised by electrolyte imbalances (hypomagnesemia), urinary enzyme excretion, acute reduction in glomerular filtration rate and in some cases, a transient increase in blood urea nitrogen and serum creatinine.

Chronic phase: It is characterized by a stable, reduced creatinine clearance with or without elevations in serum creatinine. Renal toxicity may be worsened by concomitant administration of other renal toxins including amino glycoside antibiotics.

Regimes of I.V. hydration, mannitol diuresis and 6-8 hour infusions of Cisplatin have been used to reduce the incidence and severity of nephrotoxicity.

Neurotoxicity

Generally, it may include peripheral neuropathy (glove & stocking distribution), auditory impairment (ototoxicity) & visual disturbances in colour perception (when used in high doses). Less commonly encountered other visual side effects include cortical blindness, papilloedema & retrobulbar neuritis.

Ototoxicity

Tinnitus and/or high frequency hearing loss has been observed in up to 31 % of patients treated with Cisplatin. Usually, the hearing loss occurs in the frequency range greater than 2000 Hz. but may affect the normal hearing range of 250-2000 Hz in 10-15% of cases. It is dose-dependent. Ototoxicity may be more severe in children. Hearing loss can be unilateral or bilateral and tends to become more frequent and severe with repeated doses and may not be reversible.

Careful monitoring of audiometry should be performed prior to and during the therapy.

Hematologic

Myelosuppression occurs in 25-30% patients receiving Cisplatin, but is usually not severe. The nadirs in platelet and leukocyte counts occur between days 18 and 23, but normal values have returned by day 39. At higher doses of Cisplatin, leukocytopenia and thrombocytopenia are more serious and may be accompanied by anemia.

Gastrointestinal

Cisplatin induces severe nausea and vomiting in almost all patients. Nausea and vomiting usually begin within 1-4 hrs after treatment and may persist for up to a week after treatment. Antiemetics and administration over several hours may cause less severe nausea and vomiting.

Electrolyte Disturbances

Hypomagnesaemia and hypocalcaemia may develop during Cisplatin therapy or following discontinuance of the drug. Regular monitoring of serum electrolyte levels and their replacement, where necessary, is advisable.

Anaphylactic like Reactions

Consisting principally of facial oedema, wheezing, tachycardia and hypotension have been reported in patients previously exposed to Cisplatin. The reaction may be controlled by IV adrenaline, corticosteroids and/or antihistamines.

Cardiovascular Toxicities

Bradycardia, bundle branch block, congestive heart failure and Reynaud’s phenomenon have been rarely observed.

Hyperuricaemia

It may occur in patients receiving Cisplatin and is more pronounced with doses greater than 50 mg/m², with peak levels occurring between 3-5 days after administration of the drug. Allopurinol may be used to reduce serum uric acid levels.

Other infrequently reported adverse effects include cytogenetic changes in peripheral blood lymphocytes, treatment related leukemia, acute ischaemic.

vascular events, local neurotoxicity after intra-arterial drug infusion, microscopic & biochemical damage to brain tissues, glucose intolerance and pancreatitis.

PRECAUTIONS AND WARNING^{8,9}

1. Cisplatin is a cytotoxic anticancer drug and should be used only by physicians experienced with Cancer chemotherapeutic drugs. As with other potentially cytotoxic compounds great caution should be exercised in handling the aqueous solution. The personnel carrying out these procedures should be adequately protected with clothing, gloves & eye shield. If the solution contacts the skin or mucosae, immediately wash the skin thoroughly with soap and water.
2. Cumulative and dose related renal insufficiency is the major dose limiting toxicity of Cisplatin. Renal function must return to acceptable limits before further doses are given. Measurement of BUN, serum creatinine and creatinine clearance should be taken before initiating cisplatin therapy and prior to subsequent courses, hydration is recommended to minimize nephrotoxicity.
3. Tinnitus or occasional decreased ability to hear normal conversations

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- are indications of ototoxicity, which has been frequently observed. It is, therefore, recommended that hearing function be evaluated periodically prior to and throughout treatment.
- Since myelosuppression may occur in patients treated with Cisplatin, subsequent courses of cisplatin should not be instituted until platelets are present at levels greater than100,000 cells/mm² and WBCs greater than 4000 cells/mm². Peripheral blood counts should be monitored at regular intervals for the duration of Cisplatin treatment.
 - Anaphylactic like reactions to Cisplatin have been reported. These reactions have occurred within minutes of administration to patients with prior exposure to Cisplatin and have been alleviated by administration of epinephrine, steroids and antihistamines.
 - Hypomagnesaemia and Hypocalcaemia occur with Cisplatin administration but do not appear to be dose related. Monitoring of electrolytes is therefore necessary.
 - Peripheral neuropathy, postural hypotension and seizures may occur with Cisplatin administration and appear to be more common after prolonged administration. Neurological examinations should be performed regularly.
 - Liver function should also be monitored periodically.

USE IN PREGNANCY AND LACTATION

Safe use in pregnancy has not been established. Cisplatin is mutagenic in bacteria and produces chromosome aberrations in animal cells and tissue culture. In mice, Cisplatin is teratogenic and embryotoxic. After I.V. administration, Cisplatin is rapidly distributed into tissues including breast milk. To avoid possible harmful effects, it is recommended that Cisplatin therapy should not be initiated in lactating mothers.

DOSAGE AND ADMINISTRATION^{10,11,12}

Non-seminomatous germ cen carcinoma -metastatic

Adults & Children:
Cisplatin 20 mg/m² via IV infusion as a single dose once daily for 5 days with bleomycin and etoposide. Repeat every 3 weeks for 2 cycles or more.

Ovarian Cancer -Advanced

In combination with cyclophosphamide
Adults: 50-60 mg/m² via IV infusion administered as a single dose once every 21 days

With Paclitaxel:

Adults: 75 mg/m² via IV infusion administered as a single dose once every 21 days.

Bladder Cancer -Advanced

Administered as a single agent at a dose of 50 to 70 mg/m 2 Intravenously once every 3 to 4 weeks in combination with other neoplastic agents.

a) Pre Treatment Hydration

Patients should be adequately hydrated before and for 24 hrs. after Cisplatin administration in order to induce diuresis and minimise nephrotoxicity. Hydration may be achieved either by I.V. infusion of 2 liters of 0.9% sodium chloride or dextrose saline (Dextrose 5% in one fifth normal saline (0.18% Sodium chloride injection) over a 6-12 hr. period. During the last 30 minutes of the pre treatment hydration or after the hydration, 375 ml of 20% mannitol injection may be administered via a side arm drip.

b) Preparation of Cisplatin Infusion

Cisplatin should be added to 2 liters of 0.9% sodium chloride injection or Dextrose-Saline solution.

c) Treatment

Following prehydration, Cisplatin infusion is administered over 1-2 hrs. A longer infusion time of 6-8 hrs. may decrease gastrointestinal and renal toxicities.

d) Post Treatment Hydration

It is recommended that I.V. hydration should continue after treatment with administration of 2 liters
0.9% sodium chloride IV infusion or Dextrose-saline over a period of 6-12 hours.

NOTE

Since aluminum reacts with and inactivates Cisplatin, it is important not to use needles or other equipment that contain aluminum while preparing or administering Cisplatin.

DOSE MODIFICATIONS

- Dose of Cisplatin has to be reduced in cases of:
- Low WBC, RBC, platelets (myelosuppression)
 - Renal failure
 - Severe, un controlled nausea or vomiting

OVERDOSAGE

Acute over dosage with cisplatin may result in kidney failure, liver failure, deafness, ocular toxicity, significant myelosuppression, intractable nausea and vomiting and/or neuritis. In addition, death can occur following ovedosage. No proven antidotes have been established for cisplatin overdosage. Management of overdosage should include general supportive measures to sustain the patient through any period of toxicity that may occur.

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS

Store below 30°C. Protect from light. Do not freeze.
Great care should be taken to prevent exposure of the skin to cisplatin. The use of gloves is recommended. Any cisplatin solution that comes In contact with skin or mucosae should be washed off with soap and water. Since aluminum reacts with and inactivates cisplatin, it is important not to use needles or other equipment that contain aluminium while preparing or administering KEMOPLAT (Cisplatin Injection).

SHELF LIFE

18 months.

PRESENTATION

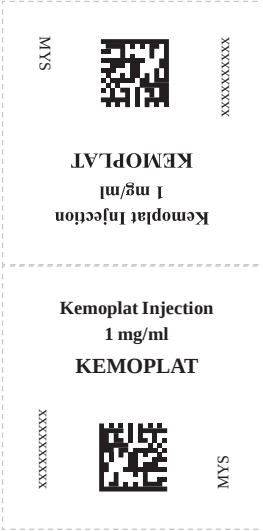
KEMOPLAT (Cisplatin Injection 1.0mg/ml) is available as 50ml single dose vial containing 50mg Cisplatin.

MANUFACTURER INFORMATION:

Manufactured in India by:
Fresenius Kabi Oncology Limited,
Village Kishanpura, Baddi,
Tehsil Nalagarh, District Solan,
Himachal Pradesh, IN 174101, India

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